Rakey | Ruoqing Yang

rakeyyang.com · rakeyyang@gmail.com · +44 7355 399713

Education

- **University of Nottingham** 2023–2024
  MSc, Human-Computer Interaction (Distinction) *Nottingham, England*
- **Ningbo University** 2018–2022
  BSc, Supply Chain and Logistics Management *Ningbo, China*
  Dissertation: Human–Machine Interaction in Logistics Warehousing

Publications

- **R. Yang** and C. Greenhalgh. "Comparing Experience Intensity of AR and VR for Contrasting Phobia Stimuli." *Virtual Reality*, 2026.
- S. Jin, **R. Yang**, W. Tong, and L. H. Lee. "AIs or Humans in Luxury Shopping? A Design Study of VR Shopping Assistants." *2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), pp. 1155–1156*, 2026.
- S. Jin, T. Bohné, **R. Yang**, W. Tong, and L. H. Lee. "Comparing Assistant Configurations in Simulated VR Luxury Shopping: A Mixed-Methods Study." *Virtual Reality*, 2026.

Research Experience

- **Comparing Experience Intensity of AR and VR for Contrasting Phobia Stimuli** 2024–2025
  MSc Dissertation · University of Nottingham *Nottingham, England*
  - Built matched VR and AR exposure environments in Unity using spider and contamination stimuli.
  - Designed and ran a headset-based AR/VR study (n=31); analysed the data using two-way repeated-measures ANOVA and reflexive thematic analysis, and wrote up the findings for publication.
  - Found that VR elicited higher experience intensity overall, while AR performed comparably and offered greater ecological validity in the spider scenario, suggesting a staged AR-then-VR approach.
- **User Perception of Different Shopping Assistants in VR Luxury Store** 2025–2026
  Research Collaboration · University of Cambridge *Cambridge, England*
  - Co-designed and built a multi-user PC-VR luxury store with four assistant modes: 3D avatar, webcam, voice, and AI.
  - Ran a comparative user study (n=20) using NASA-TLX, questionnaires, and interviews; collected and cleaned the data, and produced the figures, tables, presentations, and videos.
  - Found that human assistants received better preference ratings and identified a realism-immersion trade-off.

Projects

- **One's Own Room – Adaptive WebXR Environment** *[GitHub]*
  - Designed a personalised WebXR space that adapts atmosphere in response to users' affective states.
  - Built the space with GenAI-generated 3D worlds; integrated LLM + TTS APIs for real-time conversation.
- **AR Gallery Treasure Hunt Game** *[Video]*
  - Modelled exhibition environments and 3D content from Nottingham Lakeside Arts exhibits and venue geography.
  - Designed location-based AR for spatial exploration; task completion rate was 90% (n=15).

Professional Experience

- **AI Design Specialist** 01/2024–07/2024
  O Shaped · Internship *London, England*
  - Built AI-avatar and generative-video workflows for educational content, reducing production time by 40%.
  - Iterated avatar appearance, behaviour, and speech based on user feedback, improving clarity and usability.

Skills

XR Development: Unity (C#) - VR/AR/MR (Passthrough, Hand/Eye Tracking), WebXR (Three.js, WebGL)
Research Methods: Mixed Methods (Quant/Qual), Research-through-Design, Co-design, Experimental Design, User Studies
Data Analysis: Quant - R, SPSS, Python; Qual - Qualitative Coding, Thematic Analysis
Design: Figma, Blender, Adobe Creative Suite **Languages:** English, Mandarin, Taiwanese Hokkien

Awards & Activities

Presenter, GOSIM AI Spotlight, Paris (2026); Best AI Use Prize, RealityX AI × XR Hackathon (2026). First Prize, Cambridge EduX Hackathon (2025); Winter School - Artificial Intelligence and Machine Learning, University of Oxford (2022).